

Revised Generation Interconnection System Impact Study Report

For Queue Project Q20-1431

Abajo 138 kV
May 30, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1431. The Interconnection Customer proposes a 20.8 MW battery energy storage facility interconnecting at the Abajo 138 kV substation in the Kentucky area. The request was received on March 17, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 504 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,425,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Carbon Ceter 500/345 kV XFMR is overloaded to 114.4% of its 1,998 MVA emergency rating for the loss of the Blackrock-Shawville Station Substation 345kV. The project contributes 5.4% DFAX (1.1 MW) to this overload.

For the loss of Blackrock-Shawville Sta 345kV, the Carbon Ctr-Shawville Sta 500kV loads to 107.8% of its 2,111 MVA rating. Project DFAX: 5.4%.

For the loss of Piney Creek Project-Hilltop 500kV Line, the Shawville Sta-Orchard Rd Substation 500kV line loads to 103.6% of its 2,944 MVA rating. Project DFAX: 7.3%.

The Cortez-S 138kV is overloaded to 100.9% of its 197 MVA emergency rating for the loss of the Summit Lake Wps-Edna 138kV. The project contributes 25.0% DFAX (5.2 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Carbn Ctr 500 kV/345 kV Auto	Replace transformer with higher-rated unit	\$20,522,000	1% (\$287,000)
Carbon Ctr-Shawville Sta 500kV	Reconductor line	\$39,244,000	1% (\$549,000)
Shawville Sta-Orchard Rd Substation 500kV line	Reconductor line	\$40,050,000	1% (\$561,000)
Cortez-S 138kV	Reconductor line	\$862,000	3% (\$28,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
	Total allocated to project		\$1,425,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.